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The Co-Pricing Factor Zoo

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1 Big picture

- **Research question.** Which observable factors, if any, jointly price corporate bond returns and stock returns? More specifically, after combining the corporate bond factor zoo, the equity factor zoo, and nontradable macro-finance factors, can one find a robust stochastic discount factor (SDF) for both asset classes?
- **Main idea.** The paper treats the factor zoo as a model uncertainty problem. Instead of choosing one low-dimensional factor model, it studies every possible combination of 54 candidate factors, which gives $2^{54} \approx 18$ quadrillion possible linear SDF models.
- **Core empirical object.** The object of interest is not just whether a factor has a positive average return. The paper asks whether a factor belongs in the SDF that prices the joint cross-section of corporate bonds and stocks.
- **Main finding.** The true latent SDF appears to be *dense* in the space of observable factors. Dozens of observable factors are useful because they are noisy proxies for common underlying priced risks.
- **Most likely individual factors.** Only a small set of factors has posterior inclusion probability clearly above the prior: the bond and stock post-earnings announcement drift factors (PEADB and PEAD), idiosyncratic equity volatility (IVOL), the AAA/BAA credit spread (CREDIT), and the slope of the Treasury yield curve (YSP).
- **Aggregation beats selection.** Even though only a few factors are individually likely, selecting just those factors is not enough. A Bayesian Model Averaging SDF (BMA-SDF), which averages across all models, prices much better than low-dimensional models.
- **Bond factor surprise.** Once corporate bond returns are adjusted for their Treasury term structure component, equity and nontradable factors alone price the joint bond-stock cross-section almost as well as the full bond-stock factor zoo.
- **Economic interpretation.** Bond factors matter mainly because they price the Treasury component embedded in corporate bond returns. Net of that component, credit risk in corporate bonds is closely related to risks also spanned by stock-market and nontradable factors.
- **Investment implication.** The tradable version of the BMA-SDF delivers a strong out-of-sample trading strategy, with an annualized Sharpe ratio around 1.5 to 1.8 during 2004:07–2022:12.
- **Macro-finance takeaway.** The estimated SDF and its volatility are persistent, vary over the business cycle, spike in recessions and crisis periods, and forecast future factor returns.

2 Incremental contribution of the paper

- **From separate factor zoos to co-pricing.** Earlier work studies either equity factors or corporate bond factors. This paper asks whether the same pricing kernel can jointly explain both markets.
- **Comprehensive model space.** The paper does not test a handful of famous models. It

analyzes the entire model space generated by 54 factors: 16 tradable bond factors, 24 tradable stock factors, and 14 nontradable factors.

- **Bayesian solution to weak factors.** The method extends Bryzgalova, Huang, and Julliard’s Bayesian factor-zoo framework to heterogeneous asset classes. It is designed to remain valid when factors are weak, useless, or only noisy proxies for latent risks.
- **Distinction between factor probability and factor weight.** A factor’s posterior probability of being a fundamental source of risk is not the same as its posterior market price of risk. A factor can be unlikely as a true source of risk but still useful in the BMA–SDF because it proxies for latent risk.
- **Dense SDF evidence.** The paper challenges the common search for a small, correct factor model. The posterior median SDF dimension is about 22 factors, with very little support for models with fewer than 10 factors.
- **Treasury versus credit decomposition.** The duration-adjustment exercise shows that the apparent need for bond factors is largely driven by the Treasury term structure component of corporate bond returns.
- **Economic validation of the SDF.** The BMA–SDF is not fitted to forecast returns, yet its conditional mean and volatility predict future tradable factor returns. This provides an external validation of the cross-sectional pricing exercise.
- **Robustness.** The conclusions survive changes in priors, factor tilts, sparsity priors, removal of the most likely factors, alternative bond datasets, alternative test assets, expanded factor zoos, and different sample periods.

3 Data and measurement

3.1 Corporate bond data and returns

The baseline corporate bond sample combines ICE/BAML High Yield and Investment Grade index constituent data from 1997–2022 with Lehman Brothers Fixed Income data from 1986–1996. The paper then merges these data with Mergent FISD to obtain bond characteristics.

- **Sample period.** January 1986 to December 2022, for $T = 444$ monthly observations.
- **Baseline bond return.** The baseline return is the total corporate bond return minus the one-month risk-free rate.
- **Duration-adjusted return.** The paper also studies corporate bond returns net of the return on duration-matched U.S. Treasury portfolios. This removes the implicit Treasury term structure component.
- **Cleaning philosophy.** The paper avoids additional winsorizing, trimming, or augmenting the bond returns, to avoid mechanical biases in factor construction and strategy evaluation.

Interpretation. The paper is careful to separate credit-risk compensation from interest-rate or Treasury term structure compensation. This distinction becomes central for understanding why bond factors appear useful.

3.2 The joint factor zoo

The baseline factor zoo contains 54 candidate pricing factors.

- **Tradable corporate bond factors.** 16 bond factors, including credit risk, carry, duration, default, bond value, bond momentum, bond reversal, bond market, bond PEAD, and term structure factors.
- **Tradable stock factors.** 24 equity factors, including market, size, value, profitability, investment, momentum, reversal, liquidity, quality-minus-junk, betting-against-beta, mispricing, and stock PEAD factors.
- **Nontradable factors.** 14 macro-finance or state-variable factors, including credit spreads, yield-curve slope and level, inflation shocks and volatility, VIX changes, economic policy uncertainty, macro uncertainty, intermediary capital, liquidity, and idiosyncratic equity volatility.
- **Model count.** With 54 binary inclusion decisions, the baseline model space contains 2^{54} possible SDF specifications.
- **Extended zoo.** In robustness checks, the paper expands the candidate set to 91 factors when shorter sample windows permit additional data.

Interpretation. The factor zoo is not treated as a list of mutually exclusive theories. It is treated as a collection of noisy, overlapping proxies for latent priced risks.

3.3 In-sample test assets

The baseline in-sample cross-section for co-pricing includes 123 test assets.

- 50 corporate bond portfolios double-sorted on credit spreads and bond size, and on rating and time-to-maturity.
- 33 stock portfolios and anomalies similar to those used in the modern equity factor-zoo literature.
- 40 tradable factor portfolios, included because tradable factors in the model should be able to price excluded tradable factors.

Interpretation. Including tradable factors as test assets imposes a self-pricing discipline. A noisy proxy may price other assets but fail to price itself, and this helps the Bayesian method distinguish true risks from proxies.

3.4 Out-of-sample test assets

The out-of-sample cross-section is broad.

- 77 additional bond portfolios from sorts on value-at-risk, duration, bond value, bond book-to-market, long-term reversal, momentum, and industry.
- 77 additional stock portfolios from sorts on earnings-to-price, momentum, long-term reversal, accruals, size, equity variance, and industry.
- 29 Treasury portfolios with maturities from 2 to 30 years.

- The paper also evaluates $2^{14} - 1 = 16,383$ combinations of out-of-sample bond and stock cross-sections.

Interpretation. The out-of-sample design asks whether the SDF estimated on one rich but finite set of assets generalizes to many other bond and stock portfolios.

4 Methodology and empirical specifications

4.1 Linear SDF representation

The paper studies linear factor models for the SDF. Let R_t denote the N -vector of excess returns and let f_t denote the K -vector of candidate factors. A linear SDF is written as

$$M_t = 1 - (f_t - E[f_t])' \lambda_f, \quad (1)$$

where λ_f is the vector of market prices of risk.

The no-arbitrage pricing equation implies

$$E[M_t R_t] = 0, \quad (2)$$

so expected returns satisfy

$$\mu_R = C_f \lambda_f, \quad (3)$$

where C_f is the covariance between test-asset returns and factors. In empirical work, the cross-sectional pricing equation allows a common intercept and pricing errors:

$$\mu_R = \lambda_c \mathbf{1}_N + C_f \lambda_f + \alpha. \quad (4)$$

Interpretation. The model asks whether the covariance of each asset with candidate factors explains the asset's risk premium.

4.2 Why ordinary frequentist tests are fragile

Frequentist asset-pricing methods can fail when a factor is weak or useless.

- A **weak factor** has little covariance with asset returns or little cross-sectional dispersion in covariance.
- A **useless factor** is essentially unrelated to the returns being priced.
- Weak and useless factors can generate misleading risk-price estimates, invalid inference, and inflated model-fit statistics.
- The problem becomes severe in a large factor zoo because some weak or spurious factors will appear useful by chance.

Interpretation. Because the paper studies quadrillions of models, it needs an estimation method that controls the weak-factor problem automatically rather than relying on post-hoc screening.

4.3 Bayesian spike-and-slab solution

The paper extends the Bayesian method of Bryzgalova, Huang, and Julliard to the joint bond-stock setting. The key device is a spike-and-slab prior on factor risk prices:

$$\lambda_j \mid \gamma_j, \sigma^2 \sim \mathcal{N}(0, r(\gamma_j)\psi_j\sigma^2), \quad (5)$$

where $\gamma_j \in \{0, 1\}$ indicates whether factor j is included in the SDF.

- If $\gamma_j = 1$, the factor is in the model and receives a diffuse slab prior.
- If $\gamma_j = 0$, the factor receives a tight spike near zero.
- The shrinkage term ψ_j depends on the factor’s correlations with test assets and their cross-sectional dispersion.
- Weak factors are automatically shrunk toward zero.
- The baseline factor-inclusion prior is flat: each factor has prior inclusion probability 50%.

Interpretation. The prior is not meant to force a specific model. It regularizes the weak-factor problem while allowing the data to favor either sparse or dense SDFs.

4.4 Heterogeneous factor-class prior

A contribution of the paper is to generalize the prior so it can treat factor classes differently. Bond, stock, and nontradable factors may plausibly have different prior ability to explain the achievable Sharpe ratio.

The paper introduces a factor-class tilt parameter, often denoted κ , that changes the prior precision for classes of factors. For example, tilting the prior toward bond factors means assigning them a larger ex ante share of the squared Sharpe ratio.

Interpretation. This is useful because the paper combines heterogeneous asset classes. It can ask whether results depend on a prior belief that bond factors should matter more, or that stock factors should matter more.

4.5 Bayesian Model Averaging SDF

For each possible model γ^m , there is a corresponding SDF, M_{t,γ^m} . The BMA–SDF averages across all models using posterior model probabilities:

$$M_t^{BMA} = \sum_m M_{t,\gamma^m} \Pr(\gamma^m \mid \text{data}). \quad (6)$$

Because all models are linear, this can be rewritten as an average over factors:

$$M_t^{BMA} = 1 - \sum_{j=1}^K E[\lambda_j \mid data] (f_{j,t} - E[f_{j,t}]). \quad (7)$$

This identity is central.

- $\Pr(\gamma_j = 1 \mid data)$ tells whether the factor is likely to be a fundamental source of priced risk.
- $E[\lambda_j \mid data]$ tells how much weight the factor receives in the BMA-SDF or in a tradable portfolio that approximates the SDF.
- A factor can have low posterior inclusion probability but a large market price of risk if it is a useful noisy proxy for latent risk.

Interpretation. The paper separates *factor selection* from *factor aggregation*. The data strongly prefer aggregation.

4.6 Duration-adjusted bond returns

To separate credit and Treasury risks in corporate bonds, the paper defines duration-adjusted returns as

$$R_{i,t}^{bond} - R_{dur(i),t}^{Treasury} = (R_{i,t}^{bond} - R_{f,t}) - (R_{dur(i),t}^{Treasury} - R_{f,t}). \quad (8)$$

The first term is the ordinary corporate bond excess return. The second term is the Treasury component associated with a duration-matched Treasury portfolio.

Interpretation. If bond factors mainly price Treasury term structure risk, their importance should fall once the Treasury component is removed. This is exactly what the paper finds.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies which observable factors, or combinations of factors, are useful for pricing the cross-section of corporate bond and stock returns. It does not directly observe the true fundamental risks. Instead, it infers the latent SDF from how candidate factors covary with many test assets.

5.2 Dense SDF versus sparse models

The central empirical message is that the latent SDF is dense in the observable factor space.

- Sparse models with only a few observable factors have very low posterior support.
- The posterior median number of factors in the co-pricing SDF is around 22.
- This density does not imply an implausibly high Sharpe ratio, because many factors overlap and proxy for common risks.

Interpretation. The factor zoo is less like a set of distinct animals and more like a dense jungle of noisy proxies for a smaller set of latent risks.

5.3 Posterior probabilities versus market prices of risk

A key conceptual distinction is:

- **Posterior probability** answers: should this factor be included in a theoretical model of the true SDF?
- **Posterior market price of risk** answers: how much should this factor be weighted in a portfolio that best approximates the latent SDF?

Interpretation. A factor can be a poor candidate for a structural theory but still be a useful proxy in an empirical SDF portfolio.

5.4 Why self-pricing matters

Including tradable factors among the test assets helps detect noisy proxies.

- A noisy proxy can price other test assets because it is correlated with true risk.
- But it may fail to price itself correctly.
- The Bayesian procedure uses this self-pricing discipline to lower the posterior probability of noisy proxies while still using them in the BMA–SDF when they help recover latent risk.

Interpretation. This is why the method can recover the overall market price of risk even if the true factor is not directly observed.

5.5 Treasury component interpretation

The duration-adjustment results imply a clean economic decomposition.

- Corporate bond excess returns contain both credit-risk compensation and Treasury term structure compensation.
- Bond factors are good at pricing the Treasury component.
- Stock and nontradable factors are enough to price the credit component once Treasury risk is removed.

Interpretation. This does not mean bond markets are irrelevant. It means the bond factor zoo’s incremental value for co-pricing comes largely from interest-rate risk embedded in corporate bonds.

5.6 Limitations

- The method is still based on observable proxies, not direct observations of structural risks.
- Bayesian results depend on priors, although the paper runs many prior and robustness checks.
- The estimated SDF is linear in candidate factors.
- The model prices cross-sections of returns, not a fully specified general equilibrium economy.
- The findings are strongest for the sample and assets studied, though robustness checks are unusually extensive.

6 Tables and figures

6.1 Figure 1: Simulation evidence with useless factors and noisy proxies

Read:

- Figure 1 validates the Bayesian estimator in a controlled environment.
- The simulation includes a useful pseudo-true factor, useless factors, and noisy proxies of the true factor.
- Even when the true factor is absent, the BMA-SDF recovers the overall market price of risk by combining noisy proxies.
- Useless factors are shrunk toward zero in market-price terms.

Economic message: The method is designed for exactly the setting faced in the factor zoo: the true SDF may be latent, and the observed factors may be imperfect proxies.

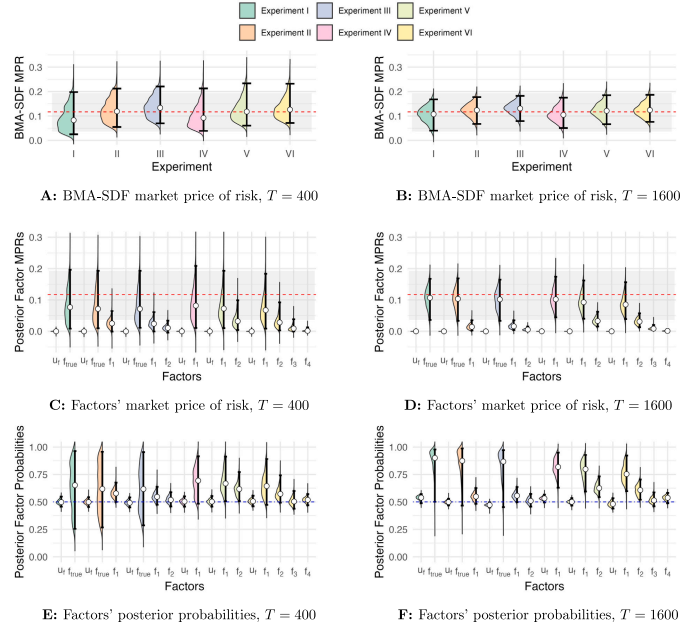


Figure 1: Simulation evidence with useless factors and noisy proxies.

6.2 Figures 2 and 3: Which factors are likely, and how dense is the SDF?

Read Figure 2:

- Figure 2 plots posterior factor inclusion probabilities across prior Sharpe-ratio values.
- The prior inclusion probability is 50%.
- PEADB, PEAD, IVOL, CREDIT, and YSP are the key factors with probabilities above the prior across the range of prior shrinkage values.

- Most other factors fall below the prior and are unlikely to be individual fundamental sources of risk.

Read Figure 3:

- The top panel shows the posterior distribution of the number of factors in the co-pricing SDF.
- The posterior median is about 22 factors, with a 95% credible interval around 15 to 29 factors.
- The bottom panel shows that the implied Sharpe ratio is not extreme despite the dense SDF.

Economic message: Only a few factors are individually likely, but the SDF is dense because many factors jointly proxy for latent priced risks.

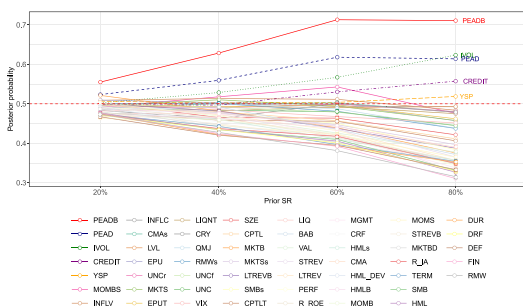


Figure 2: Posterior factor probabilities for the co-pricing factor zoo.

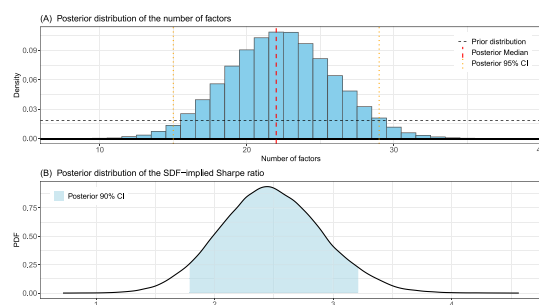


Figure 3: Posterior SDF dimensionality and implied Sharpe ratios.

6.3 Table 1 and Figure 4: Top factors and factor weights

Read Table 1:

- Table 1 reports how much of the SDF-implied Sharpe ratio is generated by the five most likely factors.
- The top five factors are important, but they do not explain the full squared Sharpe ratio of the SDF.
- This is evidence that many less likely factors still carry useful information about latent priced risk.

Read Figure 4:

- The top panel ranks factors by posterior inclusion probability.
- The bottom panel reports posterior market prices of risk.
- Some factors with probabilities below the prior still have sizeable market prices of risk.
- This illustrates the difference between being a likely fundamental risk and being a useful SDF proxy.

Economic message: Factor selection alone is too narrow. Factor aggregation is needed because observable factors contain overlapping pieces of information about the latent SDF.

Table 1

Most likely (top five) factor contribution to the SDF.

	Panel A: Co-pricing SDF				Panel B: Bond SDF				Panel C: Stock SDF			
	20%	40%	60%	80%	20%	40%	60%	80%	20%	40%	60%	80%
Total prior SR:	0.26	0.57	1.06	1.24	0.28	0.71	1.10	1.46	0.17	0.42	0.78	1.10
$E[SDF_{\text{data}}]$	0.13	0.20	0.32	0.28	0.34	0.57	0.65	0.70	0.12	0.22	0.35	0.42

Posterior mean of implied Sharpe ratios achievable with the most likely (top five) factors, $E[SDF_{\text{data}}]$, and their share of the SDF squared Sharpe ratio, $E[SDF_{\text{data}}^2]$. Panels A, B and C report results using the corresponding factor sets, for the co-pricing, bond-only, and stock-only BMA-SDFs, respectively. Top five co-pricing factors are PEAD, IVOL, PEAD, CREDIT and YSP. Top five bond factors are PEAD, CREDIT, MOMRS, YSP and IVOL. Top five stock factors are PEAD, IVOL, MKTS, CMAx and LV1. The total prior Sharpe ratio is expressed as a

Table 1: Most likely factor contribution to the SDF.

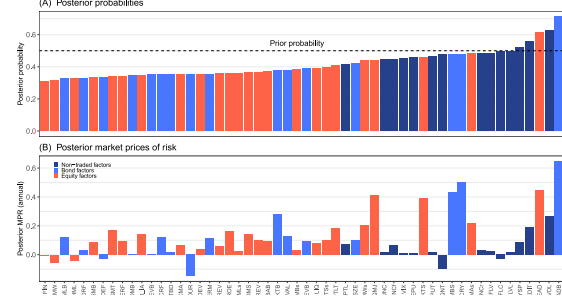


Figure 4: Posterior probabilities and posterior market prices of risk.

6.4 Tables 2 and 3: In-sample and out-of-sample pricing performance

Read Table 2:

- Table 2 compares in-sample pricing performance of BMA-SDFs and benchmark models.
- The BMA-SDF with moderate shrinkage generally dominates low-dimensional models such as CAPM, CAPMB, FF5, and HKM.
- KNS and RPPCA are stronger competitors, but the BMA-SDF is usually better.

Read Table 3:

- Table 3 repeats the comparison out-of-sample.
- The co-pricing BMA-SDF performs especially well out-of-sample for the joint bond-stock cross-section.
- The result supports the paper's claim that model averaging generalizes better than selecting a small model.

Economic message: The BMA-SDF is not just an in-sample fitting device. Its cross-sectional pricing performance remains strong on assets not used to estimate the model.

Table 2

In-sample cross-sectional asset pricing performance.

	BMA-SDF prior Sharpe ratio				CAPM	CAPMB	FF5	HKM	TOP	KNS	RPPCA
	20%	40%	60%	80%							
Panel A: Co-pricing bonds and stocks											
RMSE	0.214	0.203	0.185	0.167	0.260	0.278	0.258	0.259	0.230	0.166	0.197
MAPE	0.167	0.154	0.139	0.125	0.194	0.221	0.198	0.192	0.171	0.126	0.132
R^2_{OLS}	0.155	0.240	0.367	0.487	-0.244	-0.426	-0.233	-0.238	0.023	0.489	0.282
R^2_{OLS}	0.106	0.168	0.232	0.285	0.078	0.083	0.087	0.078	0.263	0.176	0.267
Panel B: Pricing bonds											
RMSE	0.180	0.148	0.121	0.104	0.209	0.214	0.201	0.206	0.162	0.192	0.091
MAPE	0.129	0.109	0.091	0.079	0.146	0.135	0.143	0.146	0.128	0.111	0.067
R^2_{OLS}	0.196	0.455	0.638	0.733	-0.083	-0.134	-0.006	-0.049	0.347	0.088	0.794
R^2_{OLS}	0.211	0.299	0.381	0.444	0.172	0.195	0.238	0.175	0.549	0.071	0.419
Panel C: Pricing stocks											
RMSE	0.230	0.241	0.236	0.220	0.292	0.264	0.275	0.292	0.352	0.162	0.175
MAPE	0.186	0.189	0.181	0.166	0.229	0.211	0.221	0.226	0.294	0.133	0.141
R^2_{OLS}	0.023	-0.075	-0.029	0.103	-0.570	-0.282	-0.392	-0.574	-1.288	0.515	0.433
R^2_{OLS}	0.145	0.213	0.287	0.353	0.120	0.118	0.130	0.121	0.330	0.311	0.493

Table 2: In-sample cross-sectional asset pricing performance.

Table 3

Out-of-sample cross-sectional asset pricing performance.

	BMA-SDF prior Sharpe ratio				CAPM	CAPMB	FF5	HKM	TOP	KNS	RPPCA
	20%	40%	60%	80%							
Panel A: Co-pricing bonds and stocks											
RMSE	0.114	0.102	0.095	0.090	0.224	0.154	0.139	0.223	0.171	0.160	0.153
MAPE	0.081	0.074	0.069	0.065	0.192	0.129	0.102	0.190	0.135	0.143	0.130
R^2_{OLS}	0.357	0.489	0.557	0.603	-1.478	-0.161	0.053	-1.444	-0.442	-0.268	-0.159
R^2_{OLS}	0.038	0.070	0.098	0.124	0.028	0.034	0.036	0.028	0.090	0.065	0.028
Panel B: Pricing bonds											
RMSE	0.123	0.116	0.110	0.106	0.129	0.128	0.140	0.133	0.102	0.114	0.100
MAPE	0.090	0.085	0.081	0.079	0.094	0.092	0.104	0.098	0.084	0.083	0.073
R^2_{OLS}	0.051	0.156	0.237	0.296	-0.051	-0.029	-0.231	-0.112	0.342	0.180	0.375
R^2_{OLS}	0.019	0.056	0.081	0.102	-0.004	0.024	-0.032	-0.007	0.101	0.066	0.045
Panel C: Pricing stocks											
RMSE	0.105	0.088	0.077	0.070	0.123	0.119	0.116	0.124	0.149	0.078	0.104
MAPE	0.078	0.067	0.062	0.057	0.089	0.085	0.082	0.091	0.115	0.060	0.082
R^2_{OLS}	0.298	0.508	0.620	0.683	0.032	0.099	0.136	0.019	-0.422	0.613	0.305
R^2_{OLS}	0.090	0.160	0.227	0.280	0.103	0.065	0.099	0.107	0.079	0.207	0.072

Table 3: Out-of-sample cross-sectional asset pricing performance.

6.5 Figures 5 and 6: Co-pricing versus single-zoo pricing, and stability over time

Read Figure 5:

- Figure 5 compares BMA–SDFs built from the full factor zoo, only bond factors, and only stock factors.
- For ordinary bond excess returns, the full co-pricing BMA–SDF clearly outperforms bond-only and stock-only SDFs across many out-of-sample cross-sections.
- This suggests that both factor zoos contain useful information when corporate bond returns include Treasury risk.

Read Figure 6:

- Figure 6 tracks top factors using expanding forward and backward samples.
- IVOL, PEAD, PEADB, YSP, and CREDIT appear repeatedly among the top factors.
- Factor rankings are fairly stable, despite the difficult sample period.

Economic message: The joint factor zoo matters for ordinary corporate bond excess returns, and the core factor signals are not purely a full-sample artifact.

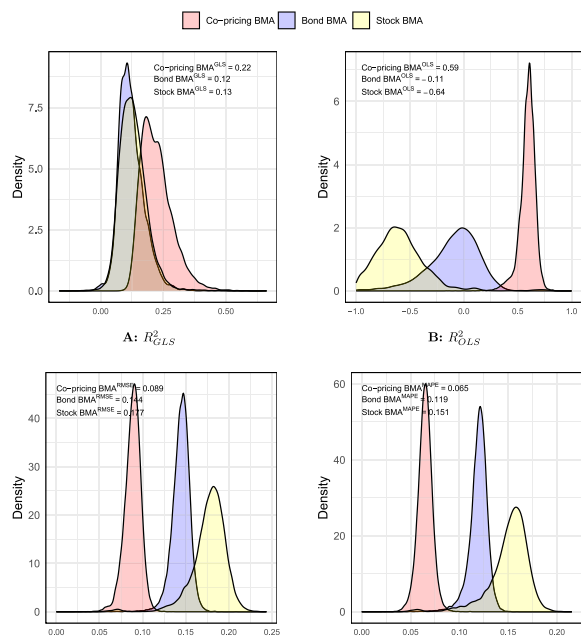


Figure 5: Pricing out-of-sample stocks and bonds with different BMA–SDFs.



Figure 6: Time-varying factor importance.

6.6 Tables 4 and 5: Risk-type decomposition

Read Table 4:

- Table 4 decomposes SDF dimensionality and Sharpe ratio by factor type.
- The co-pricing SDF is dense in nontradable factors, tradable bond factors, and tradable stock factors.
- Bond and stock factors overlap substantially: their individual Sharpe-ratio contributions sum to more than the joint contribution.

Read Table 5:

- Table 5 decomposes tradable factors into discount-rate-news and cash-flow-news groups.
- Discount-rate news is especially important for the co-pricing SDF and for the bond SDF.
- For the stock SDF, discount-rate and cash-flow news both matter.

Economic message: The priced risks are not confined to one asset class or one risk type. Both macro-finance state variables and tradable factors contribute, with discount-rate news playing an especially important role for bonds.

Table 4
BMA-SDF dimensionality and Sharpe ratio decomposition by factor type.

	Total prior SR				Total prior SR			
	20%	40%	60%	80%	20%	40%	60%	80%
Panel A: Co-pricing BMA-SDF								
	Nontradable factors				Tradable factors			
Mean	6.97	6.94	6.97	6.80	19.52	18.78	17.84	15.51
5%	4	4	4	4	14	14	13	10
95%	10	10	10	10	25	24	23	21
$\mathbb{E}[SR_f \text{data}]$	0.21	0.43	0.70	1.12	0.86	1.44	1.91	2.27
$\mathbb{E}[\frac{SR_f}{SR_n} \text{data}]$	0.08	0.11	0.15	0.23	0.94	0.93	0.90	0.84
	Tradable bond factors				Tradable stock factors			
Mean	7.85	7.50	7.21	6.32	11.67	11.28	10.63	9.19
5%	4	4	4	3	8	7	7	5
95%	11	11	11	10	16	15	15	13
$\mathbb{E}[SR_f \text{data}]$	0.56	0.95	1.28	1.51	0.66	1.13	1.50	1.77
$\mathbb{E}[\frac{SR_f}{SR_n} \text{data}]$	0.43	0.43	0.43	0.39	0.59	0.60	0.57	0.53
Panel B: Bond BMA-SDF								
	Nontradable factors				Tradable factors			
Mean	6.98	6.95	7.02	7.03	7.81	7.77	7.38	6.41
5%	4	4	4	4	5	5	4	3
95%	10	10	10	10	11	11	11	10
$\mathbb{E}[SR_f \text{data}]$	0.18	0.37	0.60	0.97	0.52	0.92	1.25	1.45
$\mathbb{E}[\frac{SR_f}{SR_n} \text{data}]$	0.15	0.18	0.22	0.33	0.86	0.83	0.78	0.66
Panel C: Stock BMA-SDF								
	Nontradable factors				Tradable factors			
Mean	6.98	7.02	6.92	7.02	11.82	11.54	11.11	9.81
5%	4	4	4	4	8	7	7	6
95%	10	10	10	10	16	16	15	14
$\mathbb{E}[SR_f \text{data}]$	0.14	0.29	0.47	0.79	0.60	1.03	1.39	1.70
$\mathbb{E}[\frac{SR_f}{SR_n} \text{data}]$	0.08	0.10	0.14	0.23	0.94	0.93	0.92	0.87

The table reports posterior means of number of factors (along with the 90% confidence intervals), implied Sharpe ratios $\mathbb{E}[SR_f | \text{data}]$, and the ratio of SR_f to the total SDF.

Table 4: BMA-SDF dimensionality and Sharpe-ratio decomposition by factor type.

Table 5
Discount rate vs. cash-flow news.

	Discount rate news				Cash-flow news			
	Total prior SR				Total prior SR			
	20%	40%	60%	80%	20%	40%	60%	80%
Panel A: Co-pricing BMA-SDF, tradable bond and stock factors								
Mean	9.81	9.60	9.28	8.20	9.71	9.18	8.56	7.31
5%	6	6	6	5	6	6	5	4
95%	14	13	13	12	13	13	12	11
$\mathbb{E}[SR_f \text{data}]$	0.65	1.19	1.70	2.10	0.60	1.06	1.45	1.77
$\mathbb{E}[\frac{SR_f}{SR_n} \text{data}]$	0.58	0.67	0.75	0.75	0.51	0.55	0.57	0.56
Panel B: Bond BMA-SDF, tradable bond factors								
Mean	4.97	5.05	4.94	4.43	2.85	2.72	2.44	1.98
5%	2	3	2	2	1	1	1	0
95%	8	8	7	7	5	5	4	4
$\mathbb{E}[SR_f \text{data}]$	0.44	0.85	1.21	1.43	0.28	0.50	0.64	0.69
$\mathbb{E}[\frac{SR_f}{SR_n} \text{data}]$	0.67	0.74	0.75	0.65	0.35	0.32	0.27	0.21
Panel C: Stock BMA-SDF, tradable stock factors								
Mean	5.01	4.91	4.79	4.38	6.81	6.63	6.31	5.43
5%	2	2	2	2	4	4	3	2
95%	8	7	7	7	10	10	9	9
$\mathbb{E}[SR_f \text{data}]$	0.37	0.73	1.11	1.48	0.47	0.83	1.16	1.44
$\mathbb{E}[\frac{SR_f}{SR_n} \text{data}]$	0.44	0.54	0.65	0.72	0.65	0.66	0.69	0.68

The table reports posterior means of number of factors (along with the 90% confidence intervals), implied Sharpe ratios $\mathbb{E}[SR_f | \text{data}]$, and the ratio of SR_f to the total SDF.

Table 5: Discount-rate versus cash-flow news.

6.7 Table 6 and Figure 7: Trading the BMA-SDF

Read Table 6:

- Table 6 forms tradable portfolios using posterior market prices of risk.
- The in-sample BMA-SDF portfolio Sharpe ratio rises with weaker shrinkage, reaching about 2.85 for the 80% prior Sharpe-ratio case.
- Out-of-sample, the BMA-SDF portfolio Sharpe ratio is about 1.46 to 1.79 across prior values.
- The 80% prior case has an out-of-sample Sharpe ratio of about 1.79.

Read Figure 7:

- Figure 7 plots cumulative out-of-sample performance from 2004:07 to 2022:12.
- The BMA-SDF strategy strongly outperforms the benchmark strategies.
- The cumulative value reaches about \$174 for the BMA-SDF versus about \$71 for RPPCA in the plotted comparison.

Economic message: Posterior market prices of risk are economically meaningful. They deliver a tradable portfolio that approximates the pricing kernel and performs well out-of-sample.

Table 6 Trading the BMA-SDF and benchmark models.													
	BMA-SDF prior Sharpe ratio				TOP γ	TOP λ	KNS	RPPCA	PF5	HKM	MKTB	MKTS	EW
	20%	40%	60%	80%									
Panel A: In-sample – 1986:01 to 2022:12 ($T = 444$)													
Mean	31.38	38.94	43.43	45.03	33.77	34.04	40.54	39.59	12.20	8.37	10.18	8.29	19.42
SR	1.99	2.46	2.75	2.85	2.14	2.15	2.57	2.51	0.77	0.53	0.64	0.52	1.23
IR	1.73	2.28	2.52	2.59	1.94	1.90	2.33	2.18	0.02	0.34	-0.47	0.29	-
Skew	0.76	0.73	0.54	0.31	0.47	0.44	0.51	0.90	-0.70	-0.65	-0.71	-0.78	-0.29
Kurt	3.55	3.98	2.47	2.90	2.53	2.54	2.98	3.07	3.41	1.91	4.68	2.22	4.63
Panel B: Out-of-sample – 2004:07 to 2022:12 ($T = 223$)													
Mean	22.72	25.73	27.17	27.90	20.59	23.41	20.36	23.01	5.90	7.12	8.22	8.71	17.15
SR	1.46	1.65	1.74	1.79	1.32	1.50	1.31	1.48	0.38	0.46	0.53	0.56	1.10
IR	0.98	1.24	1.38	1.46	1.40	1.37	0.85	1.07	-0.27	-0.26	-0.04	-0.21	-
Skew	0.30	0.04	-0.10	-0.13	-0.62	0.17	-1.19	-0.60	-1.59	-0.37	-0.93	-0.54	-1.06
Kurt	2.39	3.59	4.06	3.77	5.77	2.38	11.97	7.74	10.60	1.51	5.42	1.28	7.22

Table 6: Trading the BMA-SDF and benchmark models.

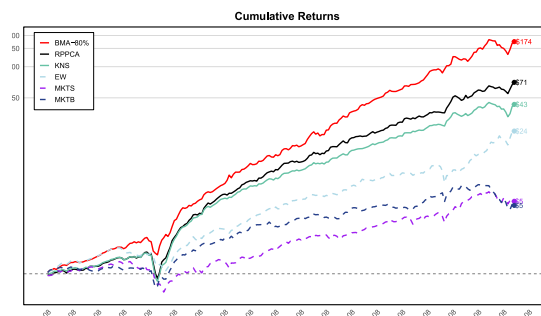


Figure 7: Out-of-sample investing in the BMA-SDF tradable portfolio.

6.8 Figures 8 and 9: Duration adjustment and the Treasury component

Read Figure 8:

- Figure 8 repeats the out-of-sample pricing comparison using duration-adjusted corporate bond returns.
- Once the Treasury component is removed, the stock-only BMA-SDF prices the joint stock and bond cross-section nearly as well as the full co-pricing BMA-SDF.
- The bond-only BMA-SDF performs much worse in this duration-adjusted setting.

Read Figure 9:

- Figure 9 directly prices the Treasury component of corporate bond returns.

- Bond factors price this component extremely well, with constrained R^2 around 97% in-sample and 92% out-of-sample.
- Stock factors fail to price the Treasury component, with constrained R^2 around 6%.

Economic message: The bond factor zoo's incremental value comes largely from pricing Treasury term structure risk embedded in corporate bond returns. Net of that component, stock and non-tradable factors are enough for co-pricing.

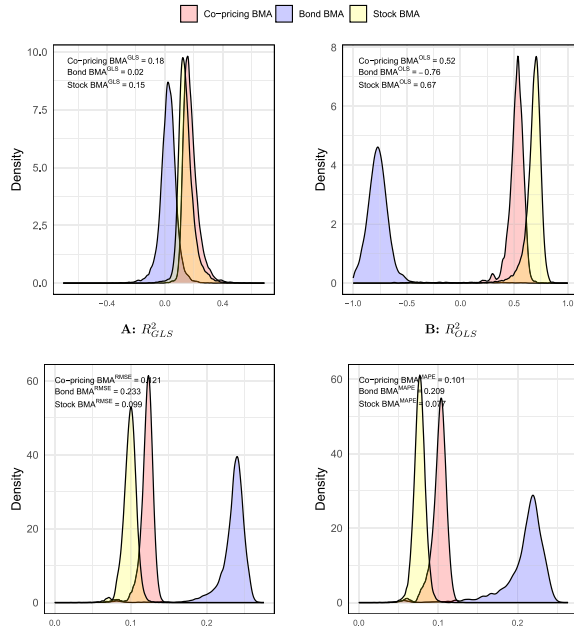


Figure 8: Pricing stock and duration-adjusted bond returns.

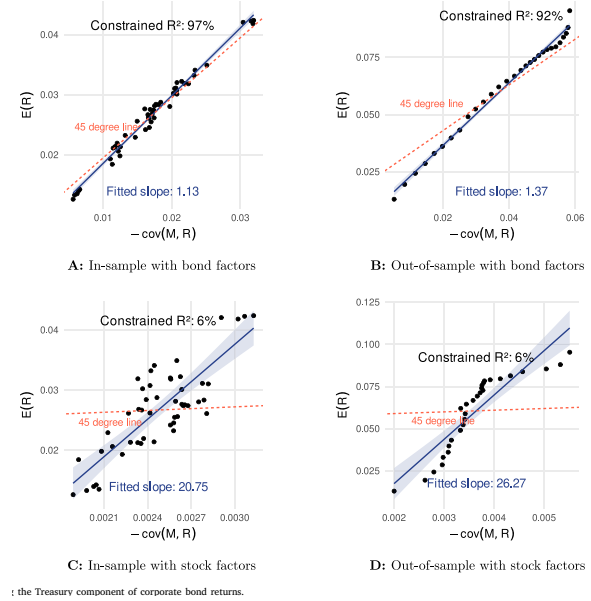


Figure 9: Pricing the Treasury component of corporate bond returns.

6.9 Figures 10 and 11: Economic properties of the co-pricing SDF

Read Figure 10:

- Figure 10 plots the BMA-SDF and its conditional mean.
- The conditional mean is persistent and exhibits clear business-cycle behavior.
- The SDF tends to move around recessions and periods of stress in a way consistent with time-varying marginal utility.

Read Figure 11:

- Figure 11 plots the conditional volatility of the BMA-SDF and compares it to CAPMB and FF5 SDF volatilities.
- The BMA-SDF volatility spikes sharply in recessions and major stress events: Black Monday, the dot-com collapse, the global financial crisis, COVID, and the Ukraine shock.

- BMA-SDF volatility is much more persistent and macro-sensitive than the benchmark model volatilities.

Economic message: The BMA-SDF behaves like a macro-finance pricing kernel: both its mean and volatility are linked to business-cycle and uncertainty states.

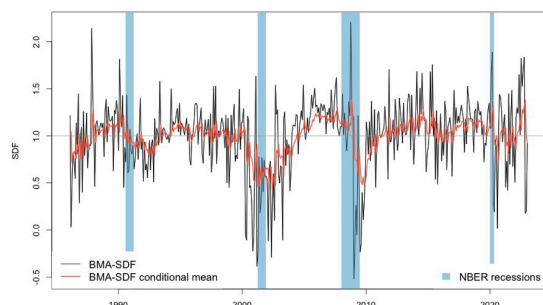


Figure 10: The co-pricing BMA-SDF and its conditional mean.

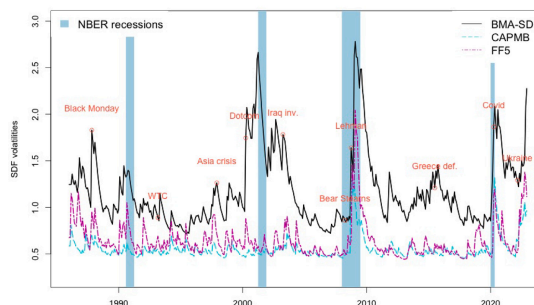


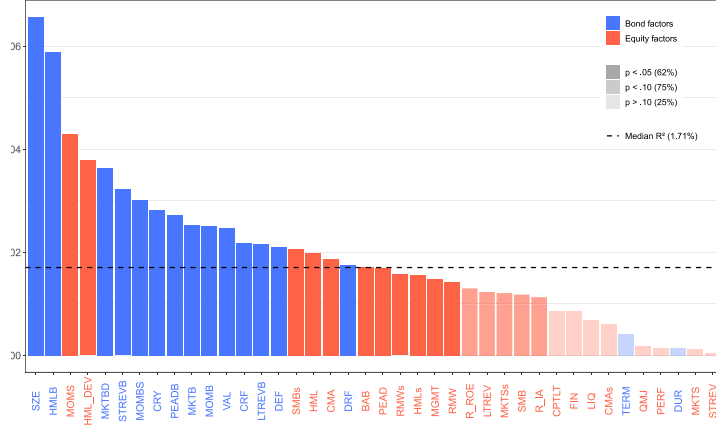
Figure 11: Conditional SDF volatilities.

6.10 Figure 12: Predictability with lagged SDF information

Read:

- Figure 12 tests whether lagged BMA-SDF information predicts future tradable factor returns.
- The regressors are the conditional variance of the BMA-SDF and the interaction of conditional variance with conditional mean.
- At the monthly horizon, many factors have statistically significant predictability.
- At the twelve-month horizon, the median R^2 is around 10%, and some factors have much higher values.

Economic message: The SDF is estimated from cross-sectional pricing, not from return predictability. Its ability to forecast future returns is therefore an important validation of the estimated pricing kernel.



A: Predictability of monthly log returns

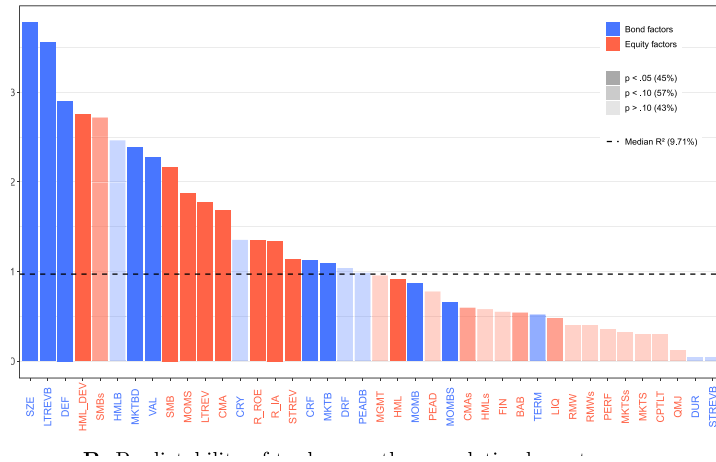


Figure 12: Predictability of tradable factors with lagged SDF information.

6.11 Appendix tables: Factor definitions and posterior probabilities

Read Appendix Table A.1:

- Table A.1 lists the 54 baseline factors, their construction, references, and data sources.
- It separates tradable corporate bond factors, tradable stock factors, and nontradable factors.
- The list clarifies that the co-pricing zoo includes not only returns-based factors but also macro-finance state variables.

Read Appendix Table C.2:

- Table C.2 reports posterior probabilities and posterior market prices of risk for every factor across prior Sharpe-ratio values.
- PEADB and PEAD stand out among tradable factors.
- IVOL, CREDIT, and YSP stand out among nontradable factors.
- Several factors with lower probabilities still have meaningful market prices of risk, reinforcing the aggregation point.

Economic message: The appendix makes the empirical machinery transparent: it documents exactly which candidate factors enter the zoo and how the posterior rankings and weights change with shrinkage.

Table A.1
List of factors for cross-sectional asset pricing. The table lists all tradable bond, stock as well as the nontradable factors used in the main paper. For each of the factors, we present their identification index (Factor ID), a description of the factor construction, and the source of the data for downloading and/or constructing the factor time series.

Factor ID	Factor name and description	Reference	Source
Panel A: Tradable corporate bond factors			
CRF	Credit risk factor. Equally-weighted average return on two 'credit portfolios': CRF_{LH} and CRF_{HL} . CRF_{LH} is the average return difference between the lowest-rating (i.e., highest credit risk) portfolio and the highest-rating (i.e., lowest credit risk) portfolio across the VARIS portfolios. CRF_{HL} is the average return difference between the lowest-rating portfolio and the highest-rating portfolio across quintiles sorted on bond short-term reversal.	Bai et al. (2019)	Open Source Bond Asset Pricing
CRV	Bond carry factor. Independent sort (5×5) to form 25 portfolios according to ratings and bond credit spreads (CS). For each rating quintile, calculate the weighted average return difference between the highest CS quintile and the lowest CS quintile. CRV is computed as the average long-short portfolio return across all rating quintiles.	Hottings et al. (2001), Houweling and Van Zundert (2017)	Open Source Bond Asset Pricing
DEF	Bond default risk factor. The difference between the return on the market portfolio of long-term corporate bond return (the Composite portfolio on the corporate bond module of Bhattacharjee) and the long-term government bond return.	Fama and French (1992) and Gebhardt et al. (2005a).	Amis Goyal website
DRF	Downside risk factor. Independent sort (5×5) to form 25 portfolios according to ratings and 90% value-at-risk (VAR90). For each rating quintile, calculate the weighted average return difference between the highest VAR5 quintile and the lowest VAR5 quintile. DRF is computed as the average long-short portfolio return across all rating quintiles.	Bai et al. (2019)	Open Source Bond Asset Pricing
DUR	Bond duration factor. Independent sort (5×5) to form 25 portfolios according to ratings and bond duration (DUR*). For each rating quintile, calculate the weighted average return difference between the highest DUR* quintile and the lowest DUR* quintile. DUR is computed as the average long-short portfolio return across all rating quintiles.	Gebhardt et al. (2005a) and Dang et al. (2023).	Open Source Bond Asset Pricing
HMLB	Bond book-to-market factor. Independent sort (2×3) to form 6 portfolios according to bond size and bond book-to-market (BBM), defined as bond principal value scaled by market value. For each size portfolio, calculate the weighted average return difference between the lowest BBM tercile and the highest BBM tercile. HMLB is computed as the average long-short portfolio return across the two size portfolios.	Bartram et al. (2025)	Open Source Bond Asset Pricing
LTREV	Bond long-term reversal factor. Dependent sort ($1 \times 3 \times 3$) to form 27 portfolios according to ratings, maturity, and the 48-12 cumulative previous bond return (LTREV*). For each rating quintile, the factor is computed as the average return differential between the portfolio with the lowest LTREV* and the one with the highest LTREV* within the rating and maturity portfolio. LTREV is computed as the average long-short portfolio return across the nine rating-maturity terciles.	Bail et al. (2021a)	Open Source Bond Asset Pricing
MKTB	Corporate Bond Market excess return. Constructed using bond returns in excess of the one-month risk-free rate of return.	Dickerson et al. (2023)	Open Source Bond Asset Pricing
MKTBD	Corporate Bond Market duration-adjusted return. Constructed using bond returns in excess of their duration-matched U.S. Treasury bond rate of return.	van Rinsbergen et al. (2025)	Open Source Bond Asset Pricing
MOBM	Bond momentum factor formed with bond momentum. Independent sort (5×5) to form 25 portfolios according to ratings and the 12-2 cumulative previous bond return (MOBM). For each rating quintile, calculate the weighted average return difference between the highest MOBM quintile and the lowest MOBM quintile. MOBM is computed as the average long-short portfolio return across all rating quintiles.	Gebhardt et al. (2005b)	Open Source Bond Asset Pricing
MOMBS	Bond momentum factor formed with equity momentum. Independent sort (5×5) to form 25 portfolios according to ratings and the 6-1 cumulative previous equity return (MOMBS). For each rating quintile, calculate the weighted average return difference between the highest MOMBS quintile and the lowest MOMBS quintile. MOMBS is computed as the average long-short portfolio return across all rating quintiles.	Hottings et al. (2001), Gebhardt et al. (2005b) and Dang et al. (2023)	Open Source Bond Asset Pricing

(continued on next page)

Factor ID	Factor name and description	Reference	Source
PEAB	Bold earnings announcement drift factor. Independent sort (2×3) to form 6 portfolios according to market equity and earnings surprise (CAR), computed according to Chan et al. (1996). For each firm size portfolio, calculate the weighted average return difference between the highest CAR terciles and the lowest CAR tercile. PEAB is computed as the average long-short portfolio return across the two firm size portfolios.	Neuwens et al. (2025)	Open Source Bond Asset Pricing
STRVB	Bond short-term reversal factor. Independent sort (5×5) to form 25 portfolios according to ratings and the prior month's bond return (REV). For each rating quintile, calculate the weighted average return difference between the lowest REV quintile and the highest REV quintile. STRVB is computed as the average long-short portfolio return across all rating quintiles.	Khang and King (2004) and Bail et al. (2021a)	Open Source Bond Asset Pricing
SIZE	Bond size factor. Dependent sort (3×3) to form 3 portfolios according to ratings and then with each rating tercile another 3 portfolios on bond size (SIZE). Bond size is defined as bond price multiplied by issue size (amount outstanding). For each rating tercile, calculate the weighted average return difference between the lowest SIZE tercile and the highest SIZE tercile. SIZE is computed as the average long-short portfolio return across all rating terciles.	Hottings et al. (2001) and Houweling and Van Zundert (2017)	Open Source Bond Asset Pricing
TERM	Bond term structure risk factor. The difference between the monthly long-term government bond return and the one-month T-Bill rate of return.	Fama and French (1992) and Gebhardt et al. (2005a).	Amis Goyal website
VAL	Bond value factor. Independent sort (2×3) to form 6 portfolios according to bond size and bond value (VAL*). VAL* is computed via cross-sectional regressions of credit spreads on ratings, maturity, and the 3-month change in credit spread. The percentage difference between the actual credit spread and the fitted (fitted) credit spread for each bond is the VAL* characteristic. For each size portfolio, calculate the weighted average return difference between the highest VAL* tercile and the lowest VAL* tercile. VAL is computed as the average long-short portfolio return across the two size portfolios.	Correia et al. (2012) and Houweling and Van Zundert (2017)	Open Source Bond Asset Pricing
Panel B: Tradable stock factors			
BAB	Betting-against-beta factor, constructed as a portfolio that holds low-beta assets, leveraged to a beta of 1, and that shorts high-beta assets, de-leveraged to a beta of 1.	Frazzini and Pedersen (2014)	AQR data library
CMA	Investment factor, constructed as a long-short portfolio of stocks sorted by their investment activity.	Fama and French (2015)	Kenneth French website
CMA4	CMA with a hedged unpriced component.	Daniel et al. (2020a)	Kent Daniel website
CPIT	The value-weighted equity return for the New York Fed's primary dealer sector but including new equity issuance.	He et al. (2017)	Zhang He website
FIN	Long-term behavioral factor, predominantly capturing the impact of share issuance and correction.	Daniel et al. (2020a)	Kent Daniel website
HML	Value factor, constructed as a long-short portfolio of stocks sorted by their book-to-market ratio.	Fama and French (1992)	Kenneth French website
HML_DEV	A version of the HML factor that relies on the current price level to sort the stocks into long and short legs.	Amew and Frazzini (2013)	AQR data library
HMLs	HML with a hedged unpriced component.	Daniel et al. (2020a)	Kent Daniel website
LQJ	Liquidity factor, constructed as a long-short portfolio of stocks sorted by their exposure to LQJNT.	Pástor and Stambaugh (2003)	Robert Stambaugh website
LTREV	Long-term reversal factor, constructed as a long-short portfolio of stocks sorted by their cumulative return across the previous 60-13 months.	Jegadeesh and Titman (2001)	Kenneth French website
MOMT	Management performance mispricing factor.	Stambaugh and Yuan (2017)	Global factor data website
MKTS	Market excess return.	Sharpe (1964) and Lintner (1965)	Kenneth French website
MKTs	Market factor with a hedged unpriced component.	Daniel et al. (2020a)	Kent Daniel website
MOMs	Momentum factor, constructed as a long-short portfolio of stocks sorted by their 12-2 months cumulative previous return.	Carhart (1997), Jegadeesh and Titman (1993)	Kenneth French website
PEAD	Short-term behavioral factor, reflecting post-earnings announcement drift.	Daniel et al. (2020a)	Kent Daniel website
PERF	Firm performance mispricing factor.	Stambaugh and Yuan (2017)	Global factor data website

(continued on next page)

Appendix Table A.1, part 1: Tradable corporate bond factors.

Appendix Table A.1, part 2: Continued bond factors and tradable stock factors.

Factor ID	Factor name and description	Reference	Source
QMJ	Quality-minus-junk factor, constructed as a long-short portfolio of stocks sorted by the combination of their safety, profitability, growth, and the quality of management practices.	Amos et al. (2019)	AQR data library
RMW	Profitability factor, constructed as a long-short portfolio of stocks sorted by their profitability.	Fama and French (2015)	Kenneth French website
RMW	RMW with a hedged unpriced component.	Daniel et al. (2020b)	Kent Daniel website
RJA	Investment factor, constructed as a long-short portfolio of stocks sorted by their investment-to-capital.	Hou et al. (2015)	Lu Zhang website
R,ROE	Profitability factor, constructed as a long-short portfolio of stocks sorted by their return on equity.	Hou et al. (2015)	Lu Zhang website
SMB	Size factor, constructed as a long-short portfolio of stocks sorted by their market cap.	Fama and French (1992)	Kenneth French website
SMBs	SMB with a hedged unpriced component.	Daniel et al. (2020b)	Kent Daniel website
STREV	Short-term reversal factor, constructed as a long-short portfolio of stocks sorted by their previous month return.	Agarwal and Titman (1993)	Kenneth French website
Panel C: Nontradable corporate bond and stock factors			
CPTL	Intermediary capital nontradable risk factor. Constructed using AR(1) innovations to the market-based capital ratio of primary dealers, scaled by the lagged capital ratio.	He et al. (2017)	Zhiguo He website
CHREIT	Bond credit risk factor. Difference between the yields of BAA and AAA indices from Moody's. Also computed with our own data as the difference between the average yield of BAA and (AAA-AAX) used bonds. See Internet Appendix B.A.1 for further computational details.	Fama and French (1993)	Asim Goyal website or FRED for AAA and BAA indices.
EPU	Economic Policy Uncertainty. First difference in the economic policy uncertainty index.	Baker et al. (2016) and Dang et al. (2023)	FRED
EPUT	Economic Tax Policy Uncertainty. First difference in the economic tax policy uncertainty index.	Baker et al. (2016) and Dang et al. (2023)	FRED
INFLC	Shocks to core inflation. Unexpected core inflation component captured by an ARMA(1,1) model. Monthly core inflation is calculated as the percentage change in the seasonally adjusted Consumer Price Index for All Urban Consumers. All from Less Food and Energy which is lagged by one-month to account for the inflation data release lag.	Baker et al. (2016) and Dang et al. (2023)	FRED
INFLV	Inflation volatility. Computed as the 6-month volatility of the unexpected inflation component captured by an ARMA(1,1) model. Monthly inflation is calculated as the percentage change in the seasonally adjusted Consumer Price Index for All Urban Consumers (CPI) which is lagged by one-month to account for the inflation data release lag.	Kang and Pfaffinger (2013)	FRED
IVOL	Idiosyncratic equity volatility factor. Cross-sectional volatility of all firms in the CRSP database in each month.	Campbell and Takaler (2003)	CRSP
LVL	Level term structure factor. Constructed as the first principal component of the one- through 30-year CRSP Fixed Term Index U.S. Treasury Bond yields.	Kojan et al. (2017)	CRSP indices
LQNT	Liquidity factor, computed as the average of individual-stock measures estimated with daily data (residual predictability, controlling for the market factor).	Piotter and Stambaugh (2003)	Robert Stambaugh website
UNC	First difference in the Macroeconomic uncertainty index.	Ludvigson et al. (2013) and Hall et al. (2013)	Sydney Ludvigson website
UNCf	First difference in the Financial economic uncertainty index.	Ludvigson et al. (2013)	Sydney Ludvigson website
VIX	First difference in the Real economic uncertainty index.	Ludvigson et al. (2015)	Sydney Ludvigson website
YSP	Slope term structure factor. Constructed as the difference in the five and one-year U.S. Treasury Bond yields.	Chang et al. (2019)	CRSP indices

Table C.2
Posterior factor probabilities and risk prices for the co-pricing factor zoo.

Factors	Factor prob., $E[r_j \text{data}]$				Price of risk, $E[\lambda_j \text{data}]$			
	Total prior Sharpe ratio				Total prior Sharpe ratio			
	20%	40%	60%	80%	20%	40%	60%	80%
PEADB	0.555	0.629	0.713	0.711	0.054	0.213	0.446	0.645
PEAD	0.523	0.559	0.618	0.614	0.035	0.138	0.297	0.449
IVOL	0.502	0.529	0.567	0.623	0.010	0.043	0.108	0.265
CREDIT	0.498	0.497	0.530	0.557	0.008	0.033	0.084	0.191
YSP	0.507	0.502	0.504	0.519	0.003	0.014	0.034	0.088
MOMBS	0.492	0.518	0.543	0.476	0.059	0.200	0.366	0.432
INFLV	0.509	0.514	0.511	0.484	0.002	0.007	0.014	0.022
INFLC	0.500	0.501	0.494	0.492	-0.001	-0.004	-0.011	-0.028
CMA	0.489	0.500	0.502	0.480	0.015	0.061	0.131	0.215
LVL	0.495	0.493	0.491	0.493	0.000	0.002	0.006	0.019
EPU	0.509	0.503	0.498	0.457	0.001	0.004	0.008	0.009
UNCf	0.494	0.490	0.499	0.480	0.001	0.004	0.012	0.032
MKT	0.496	0.510	0.494	0.458	0.055	0.173	0.289	0.391
EPUT	0.500	0.492	0.497	0.462	0.003	0.009	0.016	0.019
LQNT	0.501	0.482	0.492	0.475	-0.003	-0.013	-0.039	-0.095
CRY	0.483	0.463	0.501	0.479	0.049	0.151	0.334	0.500
QMJ	0.499	0.501	0.487	0.438	0.072	0.193	0.321	0.412
RMW	0.500	0.501	0.481	0.438	0.025	0.077	0.141	0.205
UNCf	0.499	0.492	0.479	0.446	-0.002	-0.001	0.018	0.065
UNC	0.487	0.484	0.480	0.445	-0.001	-0.000	0.005	0.014
VIX	0.482	0.485	0.468	0.452	0.000	0.002	0.005	0.010
SZE	0.502	0.465	0.464	0.421	0.006	0.026	0.061	0.104
CPTL	0.487	0.480	0.457	0.411	0.016	0.046	0.067	0.074
MKT	0.521	0.482	0.439	0.376	0.091	0.188	0.248	0.278
MKT	0.494	0.478	0.447	0.397	0.015	0.038	0.064	0.103
LTREVB	0.500	0.482	0.437	0.387	0.016	0.051	0.079	0.094
SMB	0.491	0.476	0.450	0.384	0.004	0.016	0.029	0.034
CPTL	0.478	0.459	0.456	0.406	0.023	0.068	0.130	0.186
LIQ	0.475	0.476	0.443	0.390	0.005	0.025	0.053	0.082
BAB	0.485	0.492	0.435	0.372	0.021	0.054	0.076	0.097
VAL	0.501	0.469	0.426	0.378	0.016	0.056	0.099	0.126
STREV	0.487	0.476	0.445	0.365	0.009	0.034	0.071	0.101
LTREV	0.498	0.473	0.432	0.357	0.009	0.031	0.052	0.057
PERF	0.503	0.469	0.433	0.343	0.048	0.104	0.120	0.093
R,ROE	0.490	0.465	0.416	0.357	0.049	0.103	0.135	0.159
MGMT	0.490	0.475	0.420	0.338	0.058	0.125	0.162	0.173
CRF	0.494	0.454	0.421	0.349	0.015	0.052	0.093	0.123
HML	0.478	0.461	0.411	0.357	0.004	0.011	0.021	0.026
CMA	0.469	0.464	0.421	0.351	0.028	0.063	0.077	0.063
HML DEV	0.492	0.446	0.414	0.353	0.001	0.002	0.014	0.041
HMLB	0.475	0.464	0.438	0.326	0.038	0.104	0.148	0.120
MOMB	0.472	0.459	0.424	0.346	-0.002	-0.007	-0.005	-0.003
MOMS	0.464	0.445	0.422	0.365	0.020	0.057	0.095	0.139
STREVB	0.478	0.449	0.414	0.349	0.003	0.007	0.011	0.007
MKTBD	0.487	0.442	0.403	0.351	0.014	0.029	0.029	0.015
RJA	0.473	0.437	0.418	0.349	0.034	0.079	0.120	0.140
TERM	0.474	0.443	0.397	0.354	0.027	0.058	0.085	0.116
SMB	0.476	0.434	0.410	0.331	0.010	0.044	0.079	0.086
HML	0.477	0.435	0.405	0.327	0.003	-0.016	-0.037	-0.040
DUR	0.475	0.422	0.393	0.352	0.010	-0.021	-0.081	-0.146
DRF	0.471	0.435	0.401	0.330	0.039	0.068	0.069	0.034
DEF	0.467	0.421	0.395	0.333	0.000	-0.007	-0.021	-0.030
FIN	0.476	0.424	0.392	0.311	0.034	0.035	0.015	-0.004
RMW	0.473	0.428	0.381	0.315	0.027	0.019	-0.018	-0.055

The table reports posterior probabilities, $E[r_j|\text{data}]$, and posterior means of annualized market prices of risk, $E[\lambda_j|\text{data}]$, of the 54 bond and stock factors described in Appendix A. The prior for each factor inclusion is a Beta(1, 1), yielding a prior expectation for r_j of 50%. Results are tabulated for different values of the prior Sharpe ratio, $\sqrt{E_r[S R_j^2] / \sigma^2}$, with values set to 20%, 40%, 60% and 80% of the ex post maximum Sharpe ratio of the test assets. The factors are ordered by the average posterior probability across the four levels of shrinkage. Test assets are the 83 bond and stock portfolios and 40 tradable bond and stock factors described in Section 1. The sample period is 1986:01 to 2022:12 ($T = 444$).

Appendix Table A.1, part 3: Continued stock factors and nontradable factors.

Appendix Table C.2: Posterior factor probabilities and risk prices.

7 Short summary

- The paper studies the joint pricing of corporate bonds and stocks.
- It analyzes $2^{54} \approx 18$ quadrillion SDF models built from 54 bond, stock, and nontradable factors.
- The method is Bayesian and uses spike-and-slab shrinkage to handle weak and useless factors.
- The posterior evidence rejects the idea that a small observable factor model fully prices the joint bond-stock cross-section.
- The latent SDF is dense: many factors are needed jointly, even though only a few factors are individually likely.
- The most likely individual factors are PEADB, PEAD, IVOL, CREDIT, and YSP.
- Posterior factor probabilities and posterior market prices of risk have different meanings.

- The BMA–SDF, which averages across all models, prices better than low-dimensional benchmark models in- and out-of-sample.
- A tradable BMA–SDF portfolio earns a strong out-of-sample Sharpe ratio around 1.5 to 1.8.
- Once corporate bonds are duration-adjusted, stock and nontradable factors price bonds and stocks almost as well as the full factor zoo.
- Bond factors mainly add information about the Treasury term structure component embedded in corporate bond returns.
- The BMA–SDF’s conditional mean and volatility are persistent, cyclical, and predictive of future returns.

8 Conclusion

8.1 Core conclusion

Dickerson, Julliard, and Mueller show that the joint pricing kernel for corporate bonds and stocks is dense in observable factor space. The data do not support the traditional search for one small, correct, low-dimensional factor model. Instead, many observable factors jointly proxy for a smaller set of latent priced risks.

8.2 Economic intuition

The factor zoo contains many noisy signals of common risk. Some factors, such as PEADB, PEAD, IVOL, CREDIT, and YSP, are especially likely to be sources of priced risk. But many less likely factors also help approximate the SDF because they contain partial information about latent risks. Bayesian model averaging combines these noisy signals efficiently.

8.3 Incremental contribution revisited

The paper contributes by moving from factor selection to factor aggregation. It also clarifies the bond-stock co-pricing problem: ordinary corporate bond returns include both credit and Treasury components. Bond factors are useful for the Treasury component, while equity and nontradable factors are enough for the credit component once Treasury risk is removed.

8.4 Implications

The findings imply that empirical asset pricing should be cautious about declaring one small factor model the winner. In a large factor zoo, multiple factors can proxy for the same latent risk, and excluding them can create misspecification. The paper also implies that the bond and stock markets are partially integrated through credit-related risks, but segmented through Treasury term structure risks.

8.5 Limitations

The BMA–SDF is an empirical linear approximation, not a complete structural model. The method depends on the chosen factor zoo and priors, even though the paper’s robustness checks show that

the main results are stable. The true fundamental risks remain latent and must be inferred from observable proxies.

8.6 Takeaway

The lasting takeaway is that the factor zoo should be averaged, not merely selected from. A dense BMA–SDF built from bond, stock, and nontradable factors explains corporate bond and stock risk premia better than low-dimensional models, reveals the role of Treasury risk in corporate bonds, and produces a pricing kernel with meaningful business-cycle dynamics.